



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

UNIVERSITI TEKNOLOGI MALAYSIA
FACULTY OF COMPUTING, UTM
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ASSIGNMENT 2

SECI1113 : COMPUTATIONAL MATHEMATICS

SECTION 01

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Assignment 2

$$1. (x_1, y_1) + (x_2, y_2) = (x_1 x_2, y_1 y_2) \quad c(x, y) = (cx_1, cy_1)$$

Axiom 1 :

$$(x_1, y_1) + (x_2, y_2) = (x_1 x_2, y_1 y_2)$$

is in V , (closure of addition)

Axiom 2 :

$$\begin{aligned} (x_2, y_2) + (x_1, y_1) &= (x_2 x_1, y_2 y_1) \\ &= (x_1 x_2, y_1 y_2) \\ &= (x_1, y_1) + (x_2, y_2) \end{aligned}$$

(Addition is commutative)

Axiom 3 :

$$\begin{aligned} [(x_1, y_1) + (x_2, y_2)] + (x_3, y_3) &= (x_1 x_2, y_1 y_2) + (x_3, y_3) \\ &= (x_1 x_2 x_3, y_1 y_2 y_3) \\ &= (x_1, y_1) + (x_2 x_3, y_2 y_3) \\ &= (x_1, y_1) + [(x_2, y_2) + (x_3, y_3)] \end{aligned}$$

(Addition is associative)

Axiom 6 :

$$c(x, y) = (cx_1, cy_1)$$

is in V , (closure under scalar multiplication)

Axiom 9 :

$$\begin{aligned} c[d(x, y)] &= c(dx_1, dy_1) \\ &= (cdx_1, cdy_1) \\ &= cd(x_1, y_1) \end{aligned}$$

Axiom 10 :

$$1(x, y) = (x, y)$$

2. a) $W = \{ (x, y, z) \in \mathbb{R}^3 \mid 3x = 2y \}$

$$\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \\ z_1 + z_2 \end{bmatrix} \quad (\text{subset } W \text{ is closed under addition})$$

$$\begin{aligned} 3x &= 2y \\ 3x - 2y &= 0 \end{aligned}$$

$$\begin{aligned} 3x_1 + 3x_2 - 2y_1 - 2y_2 &= 0 \\ 3(x_1 + x_2) - 2(y_1 + y_2) &= 0 \end{aligned}$$

$$c \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} cx_1 \\ cy_1 \\ cz_1 \end{bmatrix}$$

$$\begin{aligned} c(3x - 2y) &= c(0) \\ &= 0 \end{aligned}$$

(W is closed under scalar multiplication)

$\therefore W$ is subspace of $\mathbb{R}^3$

b) $\begin{bmatrix} a & b \\ a+b & 0 \\ 0 & c \end{bmatrix}$

$$\begin{bmatrix} a_1 & b_1 \\ a_1+b_1 & 0 \\ 0 & c_1 \end{bmatrix} + \begin{bmatrix} a_2 & b_2 \\ a_2+b_2 & 0 \\ 0 & c_2 \end{bmatrix} = \begin{bmatrix} a_1+a_2 & b_1+b_2 \\ a_1+a_2+b_1+b_2 & 0 \\ 0 & c_1+c_2 \end{bmatrix}$$

(W is closed under addition)

$$k \begin{bmatrix} a & b \\ a+b & 0 \\ 0 & c \end{bmatrix} = \begin{bmatrix} ka & kb \\ ka+kb & 0 \\ 0 & kc \end{bmatrix}$$

(W is closed under scalar multiplication)

3. $S = \{(4, 7, 3), (-1, 2, 6), (2, -3, 5)\}$

$$c_1 \begin{bmatrix} 4 \\ 7 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix} + c_3 \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 4 & -1 & 2 & x \\ 7 & 2 & -3 & y \\ 3 & 6 & 5 & z \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 2 & \frac{5}{3} & \frac{z}{3} \\ 7 & 2 & -3 & y \\ 4 & -1 & 2 & x \end{array} \right]$$

$$\xrightarrow{R_1 \rightarrow \frac{1}{3}R_1} \left[\begin{array}{ccc|c} 1 & 2 & \frac{5}{3} & \frac{z}{3} \\ 7 & 2 & -3 & y \\ 4 & -1 & 2 & x \end{array} \right]$$

$$R_2 \rightarrow R_2 - 7R_1$$

$$R_3 \rightarrow R_3 - 4R_1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & \frac{5}{3} & \frac{z}{3} \\ 0 & 1 & \frac{11}{3} & -\frac{1}{3}y + \frac{7}{36}z \\ 0 & -9 & -\frac{14}{3} & x - \frac{4}{3}z \end{array} \right] \xrightarrow{R_2 \rightarrow \frac{1}{11}R_2} \left[\begin{array}{ccc|c} 1 & 2 & \frac{5}{3} & \frac{z}{3} \\ 0 & 1 & \frac{11}{3} & -\frac{1}{3}y + \frac{7}{36}z \\ 0 & -9 & -\frac{14}{3} & x - \frac{4}{3}z \end{array} \right]$$

$$R_3 \rightarrow R_3 + 9R_2$$

$$\left[\begin{array}{ccc|c} 1 & 2 & \frac{5}{3} & \frac{z}{3} \\ 0 & 1 & \frac{11}{3} & -\frac{1}{3}y + \frac{7}{36}z \\ 0 & 0 & \frac{19}{3} & x - \frac{4}{3}z + 9(-\frac{1}{3}y + \frac{7}{36}z) \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & \frac{5}{3} & \frac{z}{3} \\ 0 & 1 & \frac{11}{3} & -\frac{1}{3}y + \frac{7}{36}z \\ 0 & 0 & 1 & \frac{3}{19}x - \frac{9}{76}y - \frac{5}{76}z \end{array} \right]$$

∴ Every vector can be written as a linear combination of the three given vectors.
Hence S is spanning in R^3 .

4. $2x_1 + 2x_2 + 4x_3 - 2x_4 = 0$
 $x_1 + 2x_2 + x_3 + 2x_4 = 0$
 $-x_1 + x_2 + 4x_3 - 2x_4 = 0$

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 2 & 2 & 1 & 0 \\ 4 & 1 & 4 & 0 \\ -2 & 2 & -2 & 0 \end{array} \right] \xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 2 & 2 & 1 & 0 \\ 4 & 1 & 4 & 0 \\ -2 & 2 & -2 & 0 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & 2 & 0 \\ 4 & 1 & 4 & 0 \\ -2 & 2 & -2 & 0 \end{array} \right]$$

$$\xrightarrow{R_3 \rightarrow R_3 - 4R_1} \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & 2 & 0 \\ 0 & -1 & 6 & 0 \\ 0 & 3 & -3 & 0 \end{array} \right]$$

$$\downarrow R_3 \rightarrow R_3 + R_2$$

$$\left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -9 & 0 \end{array} \right] \xrightarrow{R_3 \rightarrow \frac{1}{8}R_3} \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 8 & 0 \\ 0 & 0 & -9 & 0 \end{array} \right] \xrightarrow{R_4 \rightarrow R_4 - 3R_2} \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 8 & 0 \\ 0 & 3 & -3 & 0 \end{array} \right]$$

$$\downarrow R_4 \rightarrow R_4 + 9R_3$$

$$\left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\therefore \text{Basis} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{2} \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{2} \\ 2 \\ 1 \\ 0 \end{bmatrix} \right\}$$

The dimension is 3.

$$5. \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 4 \\ 3 & -1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 4 \\ 3 & -1 & 2 \end{bmatrix} \xrightarrow[R_3 \rightarrow R_3 - 3R_1]{R_2 \rightarrow R_2 - R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 1 \\ 0 & -7 & -7 \end{bmatrix} \xrightarrow{R_2 \rightarrow -R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \\ 0 & -7 & -7 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + 7R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & -14 \end{bmatrix}$$

$\downarrow R_3 \rightarrow \cdot \frac{1}{14} R_3$

$$\therefore \text{Rank}(B) = 3$$

The column space are :

$$C_1 = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}, C_2 = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, C_3 = \begin{bmatrix} 3 \\ 4 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

The dimension of the column space is 3

$$6. \quad B = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \quad B' = \left\{ \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$[v]_B = \begin{bmatrix} 6 \\ -1 \\ -1 \end{bmatrix}$$

a) Transition matrix B to B'

$$\left[\begin{array}{ccc|c} 2 & 0 & 1 & 1 \\ 2 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 2 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 1 \end{array} \right]$$

$$\begin{array}{l} c = \frac{1}{2} \\ b - c = 0 \\ b = \frac{1}{2} \end{array} \quad \begin{array}{l} a + \frac{1}{2}c = \frac{1}{2} \\ a + \frac{1}{4} = \frac{1}{2} \\ a = \frac{1}{4} \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{array} \right] \leftarrow \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 2 & 0 & 1 & 1 \\ 2 & 1 & 0 & -1 \\ 0 & 1 & 1 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 2 & 1 & 0 & -1 \\ 0 & 1 & 1 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & -1 & -2 \\ 0 & 1 & 1 & 1 \end{array} \right]$$

$$\begin{array}{l} f = \frac{3}{2} \\ e - f = -2 \\ e - \frac{3}{2} = -2 \\ e = -\frac{1}{2} \end{array} \quad \begin{array}{l} d + \frac{1}{2}f = \frac{1}{2} \\ d + \frac{1}{2}(\frac{3}{2}) = \frac{1}{2} \\ d = -\frac{1}{4} \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & \frac{3}{2} \end{array} \right] \leftarrow \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 2 & 3 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 2 & 0 & 1 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2 & 1 \end{array} \right]$$

$$\begin{array}{l} i = \frac{1}{2} \\ i - h = 0 \\ h = \frac{1}{2} \end{array} \quad \begin{array}{l} g + \frac{1}{2}i = 0 \\ g = -\frac{1}{4} \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{array} \right]$$

6. a) $[I]_{B'}^B = \begin{bmatrix} \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} & \frac{1}{2} \end{bmatrix}$

b) $[V]_{B'} = \begin{bmatrix} \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 6 \\ 0 \\ -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 & 6 \\ 0 & 1 & 1 & 6 \\ 0 & 1 & 2 & 1 \end{bmatrix}$

$$\begin{bmatrix} 1 & 1 & 0 & 6 \\ 0 & 1 & 1 & 6 \\ 0 & 1 & 2 & 1 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 1 & 0 & 6 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & 1 & -5 \end{bmatrix}$$

$$\xrightarrow{R_2 - R_3} \begin{bmatrix} 1 & 1 & 0 & 6 \\ 0 & 1 & 0 & 11 \\ 0 & 0 & 1 & -5 \end{bmatrix} \xrightarrow{R_1 - R_2} \begin{bmatrix} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 11 \\ 0 & 0 & 1 & -5 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -5 \\ 11 \\ -5 \end{bmatrix}$$

7. $T: M_{2,2} \rightarrow R, T(A) = a + b + c + d, A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Let $A = (a, b, c, d)$

$B = (e, f, g, h)$

$T(A+B) = T\left(\begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix}\right)$

$= (a+b+c+d) + (e+f+g+h)$

$= T(A) + T(B)$

(additivity.)

$T(kA) = T\left(k \begin{pmatrix} a & b \\ c & d \end{pmatrix}\right)$

$= ka + kb + kc + kd$

$= kT(A)$

(homogeneity)

$\therefore T$ is linear transformation

8. $B = (v_1, v_2)$

$$v_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 8 \\ 1 \end{bmatrix}; v_2 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, T\left(\begin{bmatrix} 4 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 6 \\ -3 \end{bmatrix}$$

$$c_1 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$$

$$2c_1 + 4c_2 = 10$$

$$2c_1 + 4(5 - 3c_1) = 10$$

$$2c_1 + 20 - 12c_1 = 10$$

$$-10c_1 = -10$$

$$c_1 = 1$$

$$3c_1 + c_2 = 5$$

$$c_2 = 5 - 3c_1$$

$$c_2 = 5 - 3(1)$$

$$c_2 = 2$$

$$T\begin{bmatrix} 10 \\ 5 \end{bmatrix} = 1 \begin{bmatrix} 8 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 6 \\ -3 \end{bmatrix}$$

$$= \begin{bmatrix} 8 \\ 1 \end{bmatrix} + \begin{bmatrix} 12 \\ -6 \end{bmatrix}$$

$$= \begin{bmatrix} 20 \\ -5 \end{bmatrix}$$

$$\therefore T(10, 5) = (20, -5)$$

9. $T(v) = Av \quad T: \mathbb{R}^5 \rightarrow \mathbb{R}^3$

$$T(v) = \begin{bmatrix} -1 & 3 & 2 & 1 & 4 \\ 2 & 3 & 5 & 1 & 4 \\ 2 & 0 & 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccccc|c} -1 & 3 & 2 & 1 & 4 & 0 \\ 2 & 3 & 5 & 1 & 4 & 0 \\ 2 & 0 & 2 & 2 & 0 & 0 \end{array} \right] \xrightarrow{R_1 \rightarrow -R_1} \left[\begin{array}{ccccc|c} 1 & -3 & -2 & -1 & -4 & 0 \\ 2 & 3 & 5 & 1 & 4 & 0 \\ 2 & 0 & 2 & 2 & 0 & 0 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$\downarrow$$

$$R_3 \rightarrow R_3 - 2R_1$$

v_3 & v_5 non-leading

Let $v_3 = t$

$v_5 = s$

$v_4 = 0$

$$v_2 + v_3 + \frac{1}{3}v_4 + \frac{4}{3}v_5 = 0$$

$$v_2 + t + 0 + s = 0$$

$$v_2 = -s - t$$

$$v_1 - 3v_2 - 2v_3 - v_4 - 4v_5 = 0$$

$$v_1 - 3(-s - t) - 2t - 0 - 4s = 0$$

$$v_1 + 3s + 3t - 2t - 4s = 0$$

$$v_1 - s + t = 0$$

$$v_1 = s - t$$

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{bmatrix} = \begin{bmatrix} s - t \\ -s - t \\ t \\ 0 \\ s \end{bmatrix}$$

$$\left[\begin{array}{ccccc|c} 1 & -3 & -2 & -1 & -4 & 0 \\ 0 & 9 & 9 & 3 & 12 & 0 \\ 0 & 6 & 6 & 4 & 8 & 0 \end{array} \right]$$

$$\downarrow$$

$$R_2 \rightarrow \frac{1}{9}R_2$$

$$\left[\begin{array}{ccccc|c} 1 & -3 & -2 & -1 & -4 & 0 \\ 0 & 1 & 1 & \frac{1}{3} & \frac{4}{3} & 0 \\ 0 & 6 & 6 & 4 & 8 & 0 \end{array} \right]$$

$$\downarrow$$

$$R_3 \rightarrow R_3 - 6R_2$$

$$\left[\begin{array}{ccccc|c} 1 & -3 & -2 & -1 & -4 & 0 \\ 0 & 1 & 1 & \frac{1}{3} & \frac{4}{3} & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \end{array} \right]$$

$$\downarrow$$

$$R_3 \rightarrow \frac{1}{2}R_3$$

$$\left[\begin{array}{ccccc|c} 1 & -3 & -2 & -1 & -4 & 0 \\ 0 & 1 & 1 & \frac{1}{3} & \frac{4}{3} & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{array} \right]$$

$$9. \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{bmatrix} = \begin{bmatrix} s-t \\ -s-t \\ t \\ 0 \\ s \end{bmatrix}$$

$$= s \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore \ker(T) = \{ s(1, -1, 0, 0, 1) + t(-1, -1, 1, 0, 0) \}$$

$$10. T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$B = \{v_1, v_2, v_3\}$$

$$T(v_1) = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

$$T(v_2) = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}$$

$$T(v_3) = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$

$$a) c_1 \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} -1 & 0 & -1 & -2 \\ 2 & 5 & -1 & 1 \\ 1 & 0 & 2 & 2 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 2 & 2 \\ 2 & 5 & -1 & 1 \\ -1 & 0 & -1 & -2 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 2 & 2 \\ 0 & 5 & -5 & -3 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$c_3 = 0$$

$$c_1 + 2c_3 = 2$$

$$c_2 - c_3 = -3$$

$$c_1 = 2$$

$$c_2 = -3$$

$$\downarrow \left[\begin{array}{ccc|c} 1 & 0 & 2 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$\therefore u = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} \text{ is in range of } T$$

$$b) \text{ Basis for } R(T)$$

$$R(T) = \text{span} \left\{ \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} \right\}$$